

Median of Grouped data

Daily expenses of 178 families

Daily Expenses	Number of families (fi)	Cumulative frequency (f)
0-100	38	38
100-200	37	75
200-300	32	107
300-400	28	135
400-500	43	178
Total	$n = \sum f_i = 178$	

- Take $n/2$ $178/2 = 89$
- Find the class interval having $cf > 89$ and close to 89.
- (200- 300) is the median class.
- Lower limit (l) = 200
- Class size (h) = 100
- Frequency (f) of the median class = 32
- Cumulative Frequency (C_f) of class preceding the median class= 75

Median of grouped data

- Median = $l + \left(\left(\frac{n}{2} - C_f \right) / f \right) \times h$

= $200 + \left(\left(89 - 75 \right) / 32 \right) \times 100$
= 243.75

The median daily expense is Rs.
243.75